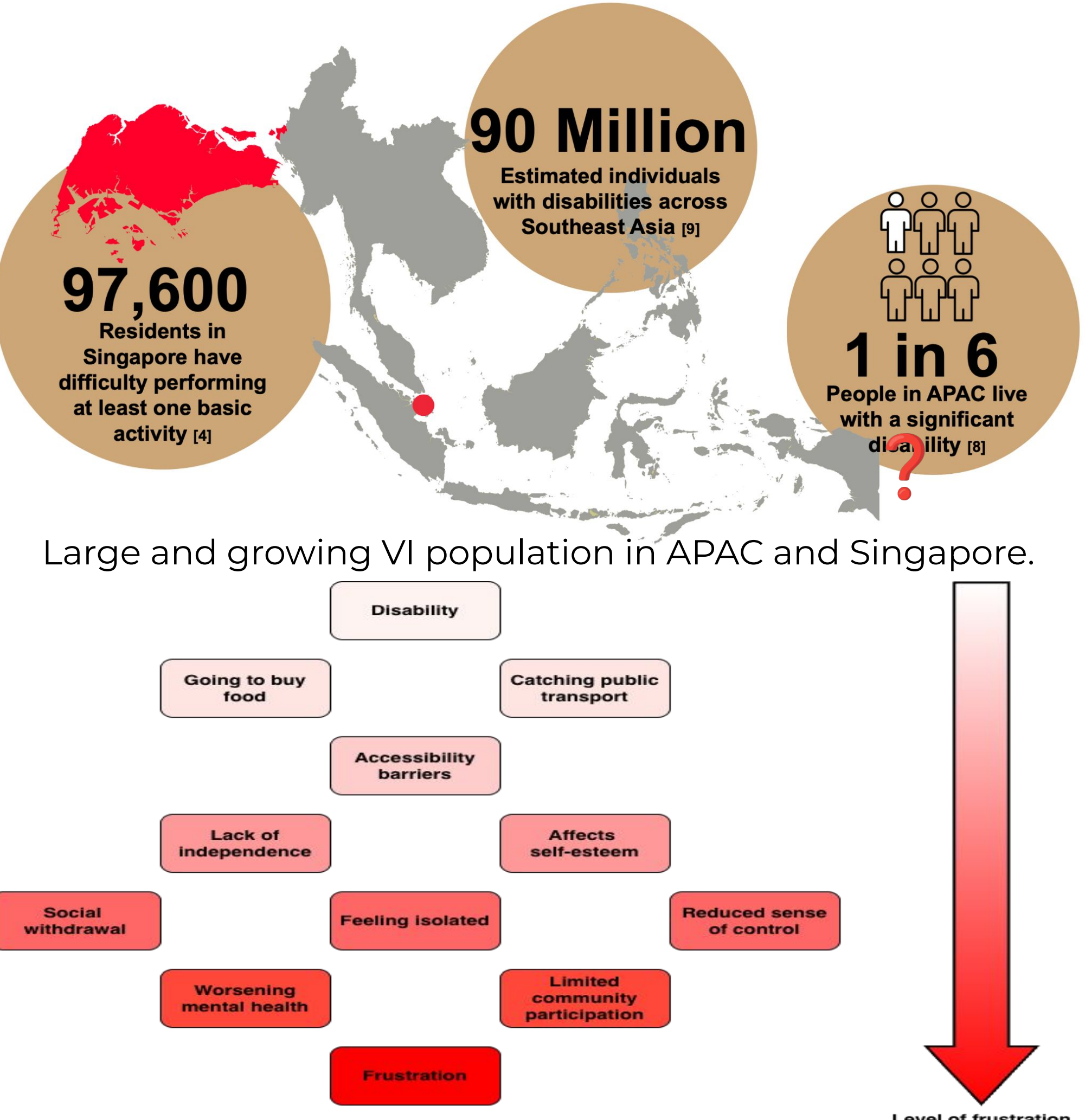
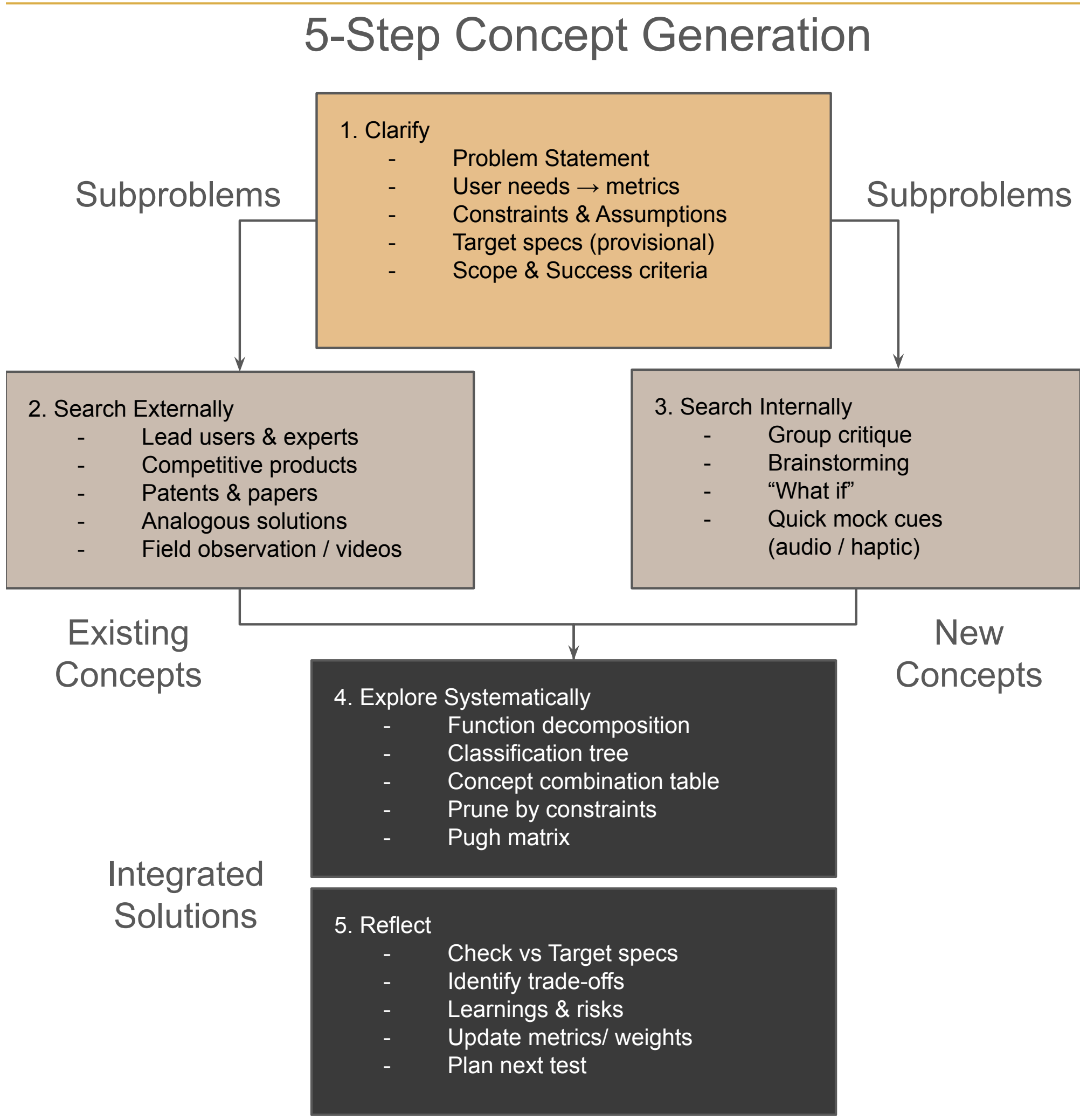


Need to address visually impaired people



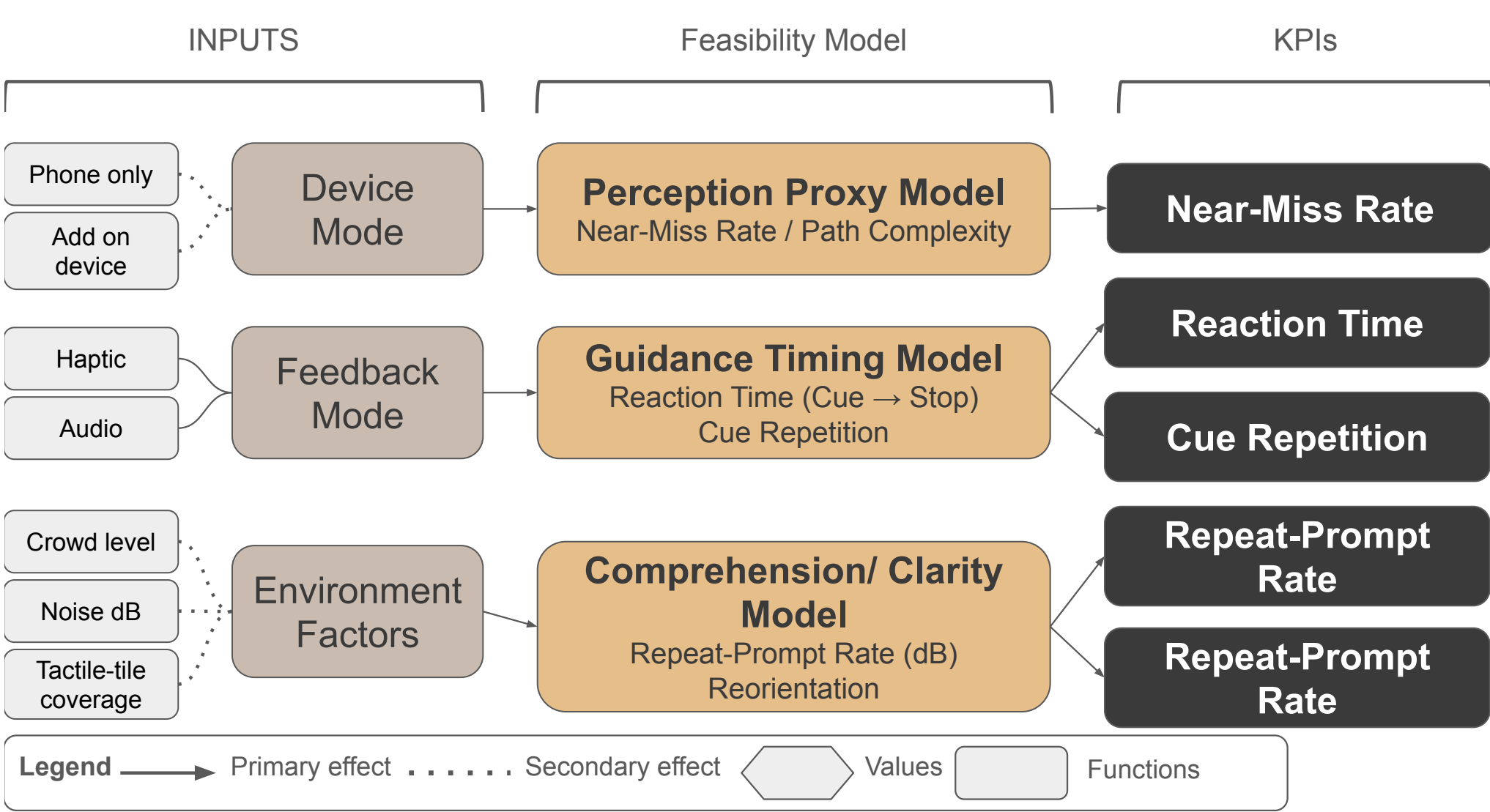
Concept Generation Process



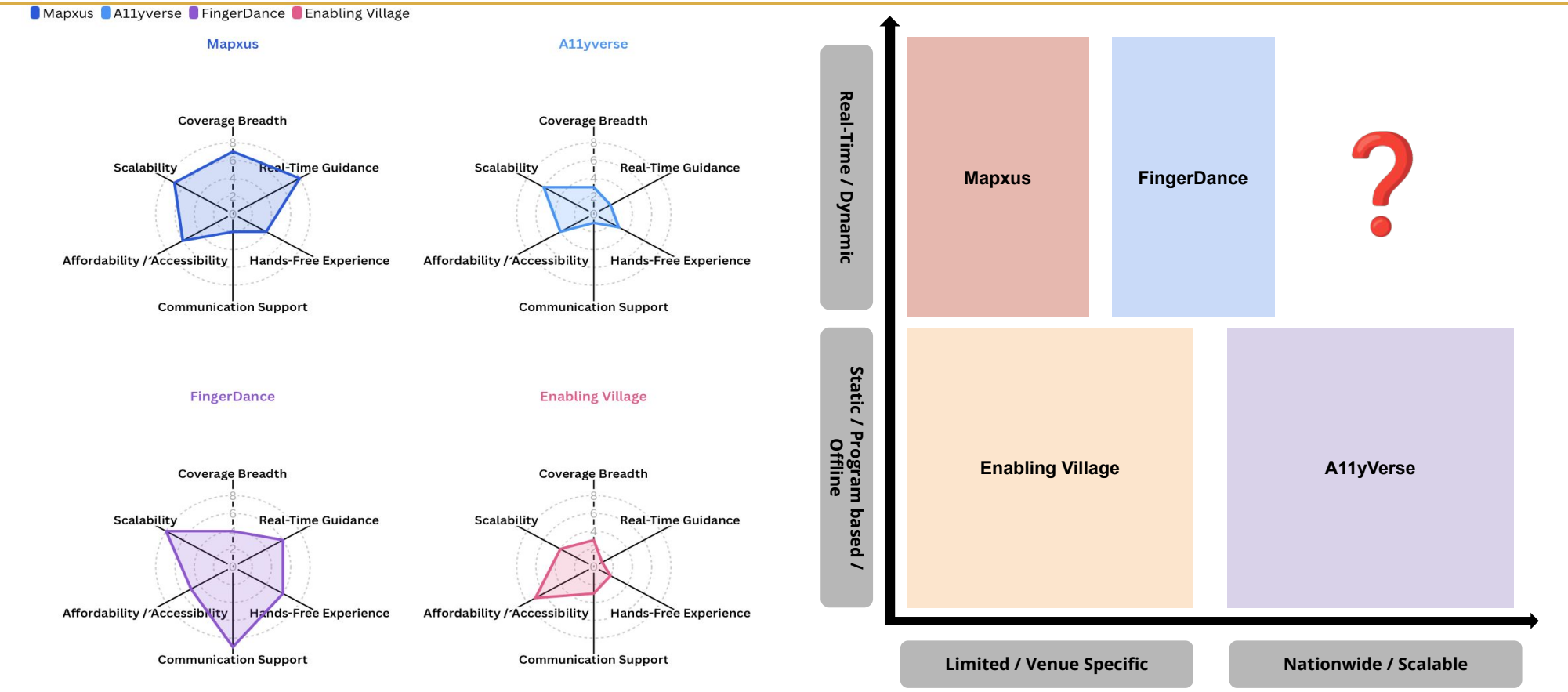
Understanding User Needs



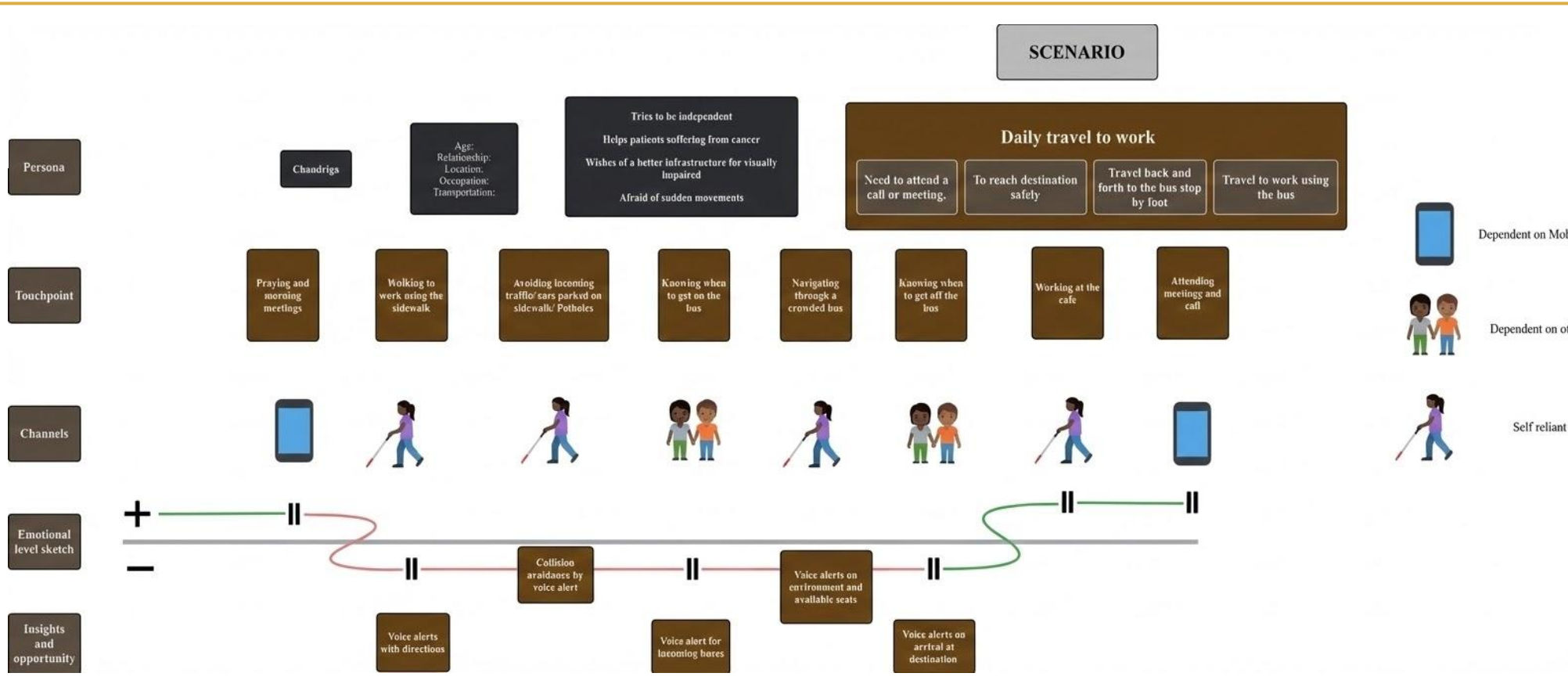
Generating KPIs through blindfold test



VI Product Market Analysis



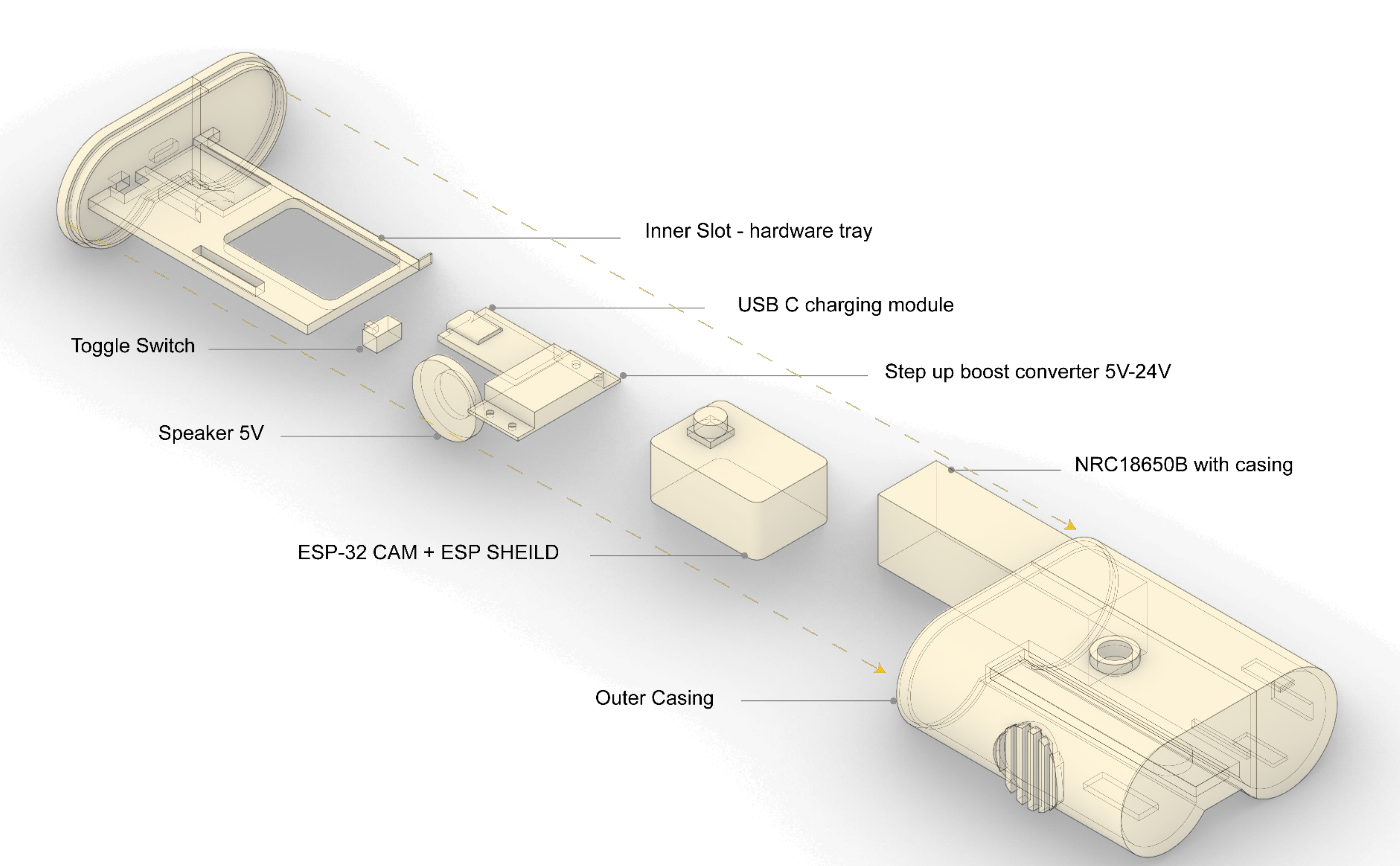
User Journey: Live Interview



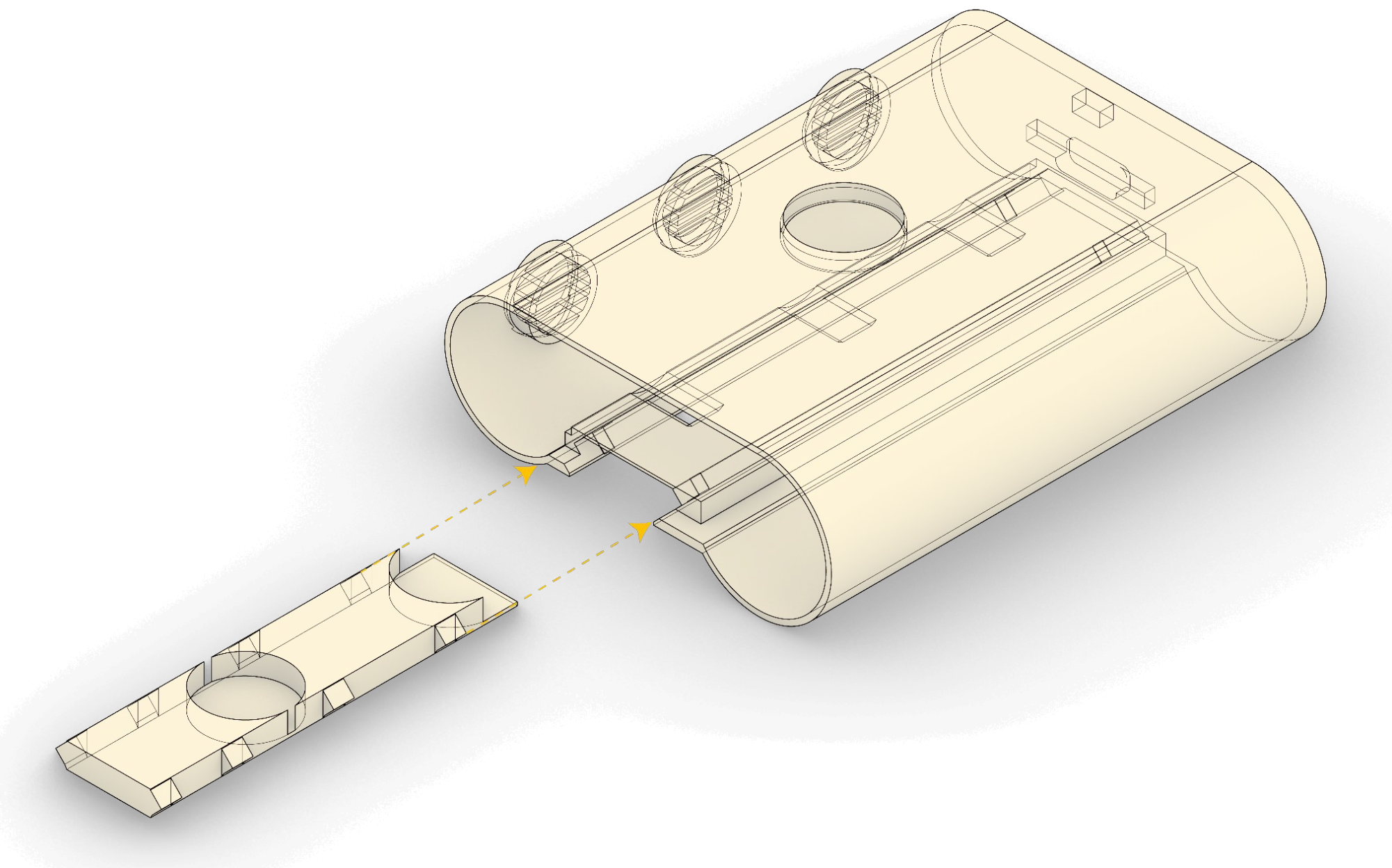
Redefining KPI through Interviews

KPI	Unit	Evidence tag	Target Calculation
Cue Reliability	% of cues delivered successfully	"Machine cues feel fast because help from humans take longer." (AiSee co-founder)	Successful cue delivery % 98% ≤ Ideal 95% ≤ Marginal
Cue Clarity (Audio Comprehension)	% of cues understood in first attempt	"Announcements down out the sound around." (Mrs. Chandrikamma)	Understood in first cue % 95% ≤ Ideal 90% ≤ Marginal
Repeat-Prompt Rate (updated weighting)	% of cues requiring repetition	"If cues come late, I freeze for a beat." (Jaavid)	Cues requiring repetition % Ideal ≤ 3-5 % Marginal ≤ 10%
User Confidence Proxy (Reorientation Count)	Reorientation attempts / 10 mins	"I don't want to tap a screen while moving." (Mrs. Chandrikamma)	Reorientation Attempts / 10mins Ideal ≤ baseline - 1 Marginal ≤ baseline
Context Robustness Score	% of correct cues in noisy / crowded settings	"Announcement down out the sound." (Mrs. Chandrikamma)	Accuracy % under noise / crowd Ideal ≤ 90% 90% ≤ Marginal
User Comfort	Average comfort ratings from users (Likert 1 → 5)	"AiSee device was extremely uncomfortable because there is only one size." (Gunhyun)	Comfort rating (1 - 5) 4.5 ≤ Ideal 4.0 ≤ Marginal

How might we provide visually impaired individuals with reliable and confidence-building navigation cues that feel faster and more dependable than seeking human assistance?



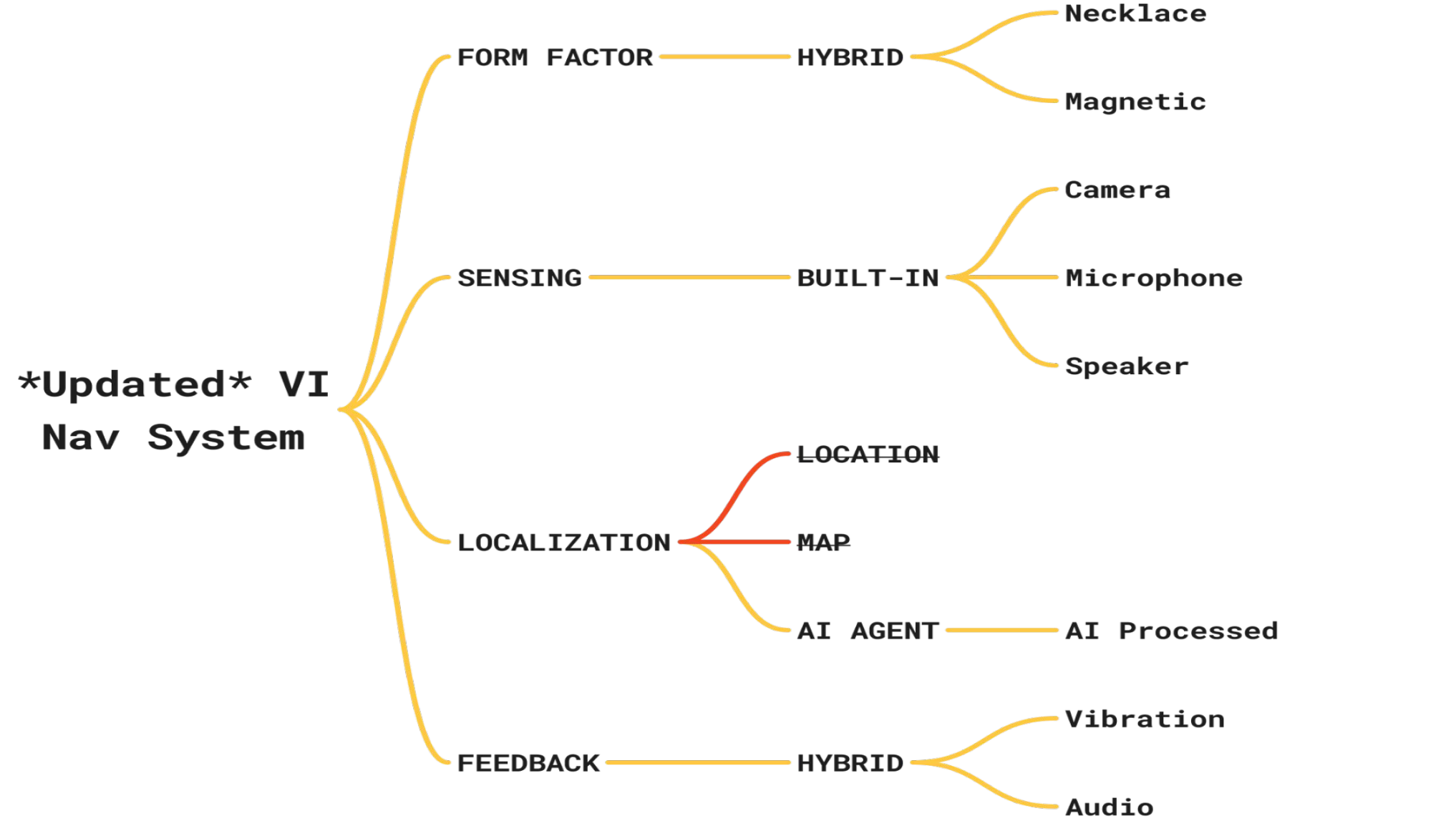
Snap Fit Mechanism - Magnet



Function / Cost BOM

Category	Recommended Part	Why It's Good
Main MCU + Camera	ESP32-CAM (OV2640)	Cheap, small, built-in Wi-Fi, perfect for snapshot workflow
Camera Ribbon	Standard 24-pin OV2640 ribbon cable	Lets you reposition the camera
USB Programmer	FTDI USB-to-Serial adapter (5V/3.3V switch)	Required to flash ESP32-CAM
Battery	3.7V LiPo 1000-2000 mAh (flat type)	Good size for wearables, lightweight
Battery Charging	TP4056 USB-C charging module (with protection)	Modern connector, safe LiPo charging
Buttons	6×6 mm toggle switch	Simple for tap and double-tap
Vibration Feedback	3-5V coin vibration motor	Good for blind-friendly feedback
Speaker	8Ω 0.5W mini speaker	Works for short TTS playback
Microphone	MAX9814 mic module	Auto-gain, good for voice memos

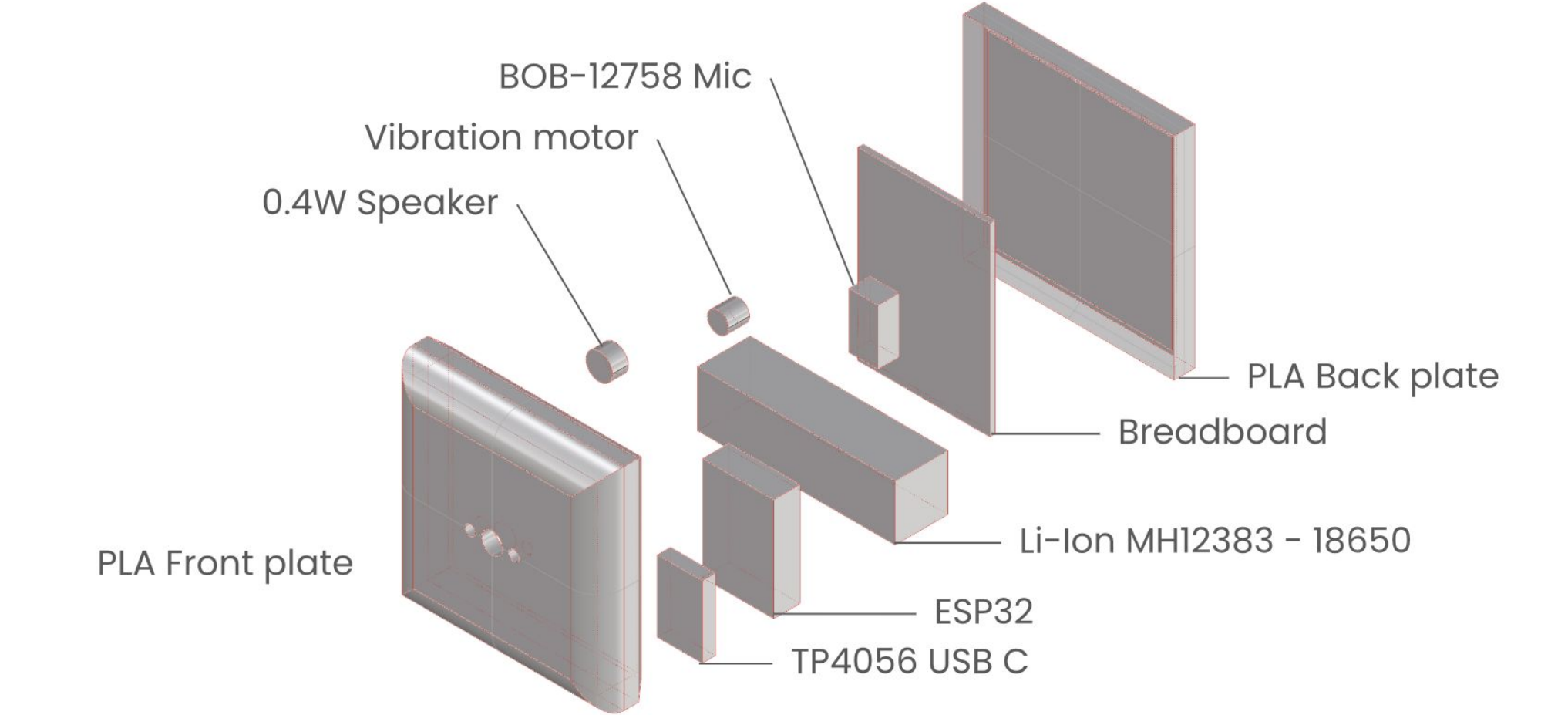
Updating Concept Fragments



Function / Cost Analysis

Problem	AiSee	Our Solution	Impact
Wearability	Ambiguous orientation	Necklace = one obvious way	Intuitive use ✓
Vision obstruction	Hair blocks side camera	Chest-mounted camera = clear view	Reliable ✓
Heat safety	Raspberry Pi causes device to heats up	ESP32: cooler + energy efficient	Safer ✓
Affordability	\$500+	<\$100 per unit	Cost efficient ✓

Iteration v.01 - Axonometric Diagram



Rapid Prototyping using GenAI

